Answer any 4 questions from 1 to 5. Each carries 1 score. **(4 × 1 = 4)**

1. The number of moles of solute present per one litre of the solution is called _____.

   (A) Molarity (B) Molality

   (C) Normality (D) Mole fraction

2. The atomic number of element with IUPAC name 'Ununquadium' is _____.

   (A) 112 (B) 110

   (C) 111 (D) 114

3. A molecule of the type AB_3E_2 has 3 bond pairs of electrons and 2 lone pairs of electrons. The most stable structure of this molecule is _____.

   (A) T-shape (B) Bent

   (C) Square planar (D) Square pyramid

4. The species that can form both conjugate acid and conjugate base among the following is :

   (A) H_2O (B) BF_3

   (C) $AlCl_3$ (D) CO_2

5. Identify the functional group with +R electron displacement effect from the following :

   $-CN, -COOH, -NO_2, -NH_2$

Answer any 8 questions from 6 to 15. Each carries 2 scores. **(8 × 2 = 16)**

6. How many grams of oxygen are required for the complete combustion of 5.6 L of CO at STP ?

7. Calculate the wave number ($\tilde{\nu}$) of the first line of the Lyman series in the hydrogen spectrum. ($R_H = 1.097 \times 10^7\ m^{-1}$)

8. Give any two postulates of Rutherford's nuclear model of an atom.

9. Draw the potential energy curve for the formation of a hydrogen molecule on the basis of inter-nuclear distance between the hydrogen atoms.

10. What is meant by entropy of a system ? What is the change in entropy when gas condenses to a liquid ?

11. (i) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

Write the expression for K_p in terms of partial pressures.

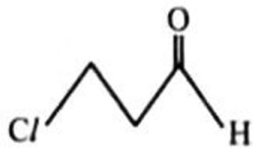
(ii) What happens to the position of equilibrium if pressure is increased for a reaction involving gases ?

12. What is disproportionation reaction ? Give example.

13. Give the IUPAC names of the following compounds :

(i) $CH_3 - CH_2 - CH = CH - CH_2 - COOH$

(ii)



14. Draw the Newman Projections of the eclipsed and staggered conformations of ethane molecule.

15. A one molecule of alkene on ozonolysis gives two molecules of ethanal. Identify the alkene. Write the chemical equation.

Answer any 8 questions from 16 to 26. Each carries 3 scores. (8 × 3 = 24)

16. A compound contains 40% carbon, 6.7% hydrogen and 53.3% oxygen by mass. Its molar mass is 180 g/mol. Find Empirical formula and molecular formula of the compound.

17. (i) The number of radial nodes of 5p orbital is _____.

(A) 1 (B) 2

(C) 3 (D) 4

(ii) Give the names of any four series of spectral lines of atomic hydrogen in the electromagnetic spectrum.

18. (i) Write the general electronic configuration of d-block elements.

(ii) A cation is smaller than corresponding neutral atom, while anion is larger. Give reason.

19. Give reason :

(i) Ionisation enthalpy of Nitrogen is greater than that of Oxygen.

(ii) Electron gain enthalpy of fluorine is lower than that of *Cl*.

(iii) Properties of first member of a group differs from those the rest of the other members of the same group.

20. (i) Dipole moment of BF_3 is zero. Why ?

(ii) Explain the shape and bond angle of NH_3 molecule using VSEPR theory.

21. Calculate the standard enthalpy of formation of ethane from the following data :

(a) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta_r H^0 = -393.5$ kJ mol^{-1}

(b) $H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow H_2O_{(l)}$; $\Delta_r H^0 = -285.8$ kJ mol^{-1}

(c) $C_2H_{6(g)} + \frac{7}{2} O_2 \rightarrow 2CO_{2(g)} + 3H_2O_{(l)}$; $\Delta_r H^0 = -1560$ kJ mol^{-1}

22. (i) Explain Bronsted-Lowry concept of acids and bases. (1)

(ii) Define Buffer Solutions and write one example each for acidic and basic buffer. (2)

23. Balance the following redox equation in acidic medium by half reaction method :

$$Fe^{2+}_{(aq)} + Cr_2O^{2-}_{7(aq)} \rightarrow Fe^{3+}_{(aq)} + Cr^{3+}_{(aq)}$$

24. (i) How will you detect the presence of nitrogen in an organic compound by Lassaigne's test ? (2)

(ii) What are nucleophiles ? Give one example. (1)

25. (i) Consider the reaction given below :

$$CH_3 - CH = CH_2 + HBr \rightarrow CH_3 - CHBr - CH_3 + CH_3 - CH_2 - CH_2Br$$

Which is the major product obtained ? (1)

(ii) Complete the following reactions :

(a) $CH_3 - Br + 2Na + Br\text{-}CH_3 \xrightarrow{\text{Dry ether}}$

(b) $3CH \equiv CH \xrightarrow{\text{Red hot iron tube / 873 K}}$ (1+1=2)

26. (i) Cycloheptatrienyl cation is given below :



Is it aromatic ? Justify the answer. (2)

(ii) Give the product formed when benzene reacts with CH_3Cl in the presence of $AlCl_3$. (1)

Answer any 4 questions from 27 to 31. Each carries 4 scores. **(4 × 4 = 16)**

27. (i) State Heisenberg's Uncertainty Principle. Give its mathematical expression. **(2)**

(ii) Quantum numbers give the address of an electron. Explain any two quantum numbers. **(2)**

28. (i) Explain hybridisation and structure of PCl_5 molecule. **(2)**

(ii) Write the molecular orbital configuration of N_2. Calculate its bond order and predict its magnetic behaviour. **(2)**

29. (i) The spontaneity of a process is expressed in terms of a change in Gibbs energy.

(a) How Gibbs energy is related to the enthalpy and entropy of a system ? **(1)**

(b) How is it useful in predicting the feasibility of a process ? **(1)**

(ii) State first law of thermodynamics. Write its mathematical expression. **(2)**

30. (i) Out of the following, which can produce an acidic solution in water ?

(CH_3COONa, NH_4Cl, CH_3COONH_4, $NaCl$) **(1)**

(ii) What is the ionic product of water ? **(1)**

(iii) Calculate the pH of a solution having H^+ ion concentration 3.8×10^{-3} M. **(2)**

31. (i) Differentiate chain isomerism from position isomerism. **(2)**

(ii) Explain the principles of the following techniques used in the purification of organic compound :

(a) Sublimation **(1)**

(b) Crystallization **(1)**
